Missing fossils as evidence: what the rocks can say about the age of land plants

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Abstract

Molecular clocks date the ancestor of living land plants from the Cambrian to the Late Ordovician, partly through different fossil maximum ages. We examined three maxima: 515.5 and 459.1 million years ago (Ma) for living land plants, and 431.2 Ma for living vascular plants. The 515.5 Ma maximum rests on a qualitative preservation argument. None of 28 Ordovician samples from Libyan boreholes reports three-rayed (trilete) spores, although 18 fall in stages in which such spores occur in Saudi Arabia and Britain: a spore absence can be local. At a steady finding rate, the 95% confidence limit on the vascular-plant record's start is 428.8 Ma; counting fewer independent finds gives 432.4–434.05 Ma. Vascular-plant ancestors at 444–452 Ma fit if fossils were found 8–12 times less often before the oldest find than after it (2.4–4.5 with fewer finds), with recognisable bodies appearing with the ancestor, or, at a steady rate, if they appeared at least 15–23 Myr later. Missing fossils allow older ancestors if early plants were found less often or became recognisable later, and these records state how much less often.

Keywords: molecular clock; fossil calibration; land plants; vascular plants; stratigraphic confidence interval; Ordovician–Silurian

Introduction

Molecular clocks estimate when lineages diverged by combining DNA differences with fossil limits on the ages of selected ancestors. A crown group comprises the most recent ancestor shared by its living members and all its descendants, including extinct lineages. Clocks date these crown ancestors; extinct lineages outside the crown may have reached land earlier. For the ancestor of all living land plants (crown embryophytes), Morris et al. [1] obtained 515–474 Ma across seven tree topologies and Harris et al. [2] 515–493 Ma. Hooper et al. [3] reanalysed Harris et al.'s DNA data with revised fossil calibrations and obtained 457–450 Ma, a mean 55 Myr younger. For the ancestor of living vascular plants (crown tracheophytes: clubmosses and their relatives, ferns and seed plants), the three studies give 451–419, 452–447 and 432–428 Ma (figure 1); Su et al. [4] obtained older land-plant ages still, under a Proterozoic maximum. The dates bear on the order in which plant characters were assembled, and on how early plants changed productivity, weathering and climate [5–8].

A fossil sets a minimum age: its lineage existed by the age of its oldest secure member. A maximum based on absence adds an argument that older rocks would probably have preserved and yielded recognisable remains had the lineage existed [9]. How calibrations are specified can strongly shape estimated dates, and with large sequence datasets they can dominate the uncertainty in absolute ages [10,11]. A soft maximum permits older ages through a specified tail of the calibration prior [12]. Whether absence can set a maximum is contested [13,14], and best practice asks each calibration to justify its maximum from the relevant geological, geographical and taphonomic record [15].

Linking a fossil to an ancestor crosses two intervals: from divergence to the first recognisable character, and from that character to its first recorded fossil [16]. Both are uncertain for early land plants. Dispersed spores appear tens of millions of years before branched plant bodies [17–19], terrestrial deposits are scarce before the latest Silurian, at least in western Europe [5], and the oldest parent plants known in detail are minute [20].

Three maxima divide the current clocks (table 1). Morris et al. set 515.5 Ma for living land plants, arguing that older marine rocks rich in algal cysts would also have yielded land-plant spores [1,14]. Hooper et al. set 459.1 Ma from the oldest possible trilete (three-rayed) spores, in the Borenshult-1 core of southern Sweden [21], and 431.2 Ma for living vascular plants from Irish *Cooksonia* [22]. The three therefore rest on a preservation analogy, on dispersed spores and on plant bodies, each with its own route into the record, and each needs its own test.

We ask three questions of the published record. What evidence supports each maximum, and what links connect it to the dated ancestor? What can the absence of spores from one Libyan basin show? How much less often would early vascular-plant fossils have to be found, or how long after the ancestor would recognisable bodies have to appear, for older dates to fit? Established theory sets an expectation for each. An argument from absence needs documented chances for remains to be delivered, preserved and found [15,16]. Where older and younger rocks were sampled comparably, each further sample reporting no target spores tightens the limit on a high, steady spore supply. With the oldest find and the time window fixed, more independent finds bring the limit on a record's start closer to that find, and a lower finding rate before it pushes the limit back [23,24]. The audit identifies what each maximum assumes, the Libyan record shows how far a spore absence reaches, and the body-fossil models describe the finding rates or delays that would fit older vascular-plant dates.

Table 1. The three maxima examined: the evidence each rests on, the test the published record allows, and the links a clock must still assume. GTS2012, Geologic Time Scale 2012.

Maximum	Set by	Evidence	Test available	Links assumed
515.5 Ma, living land plants	Morris et al.	Base of the *Olenellus* Biozone, below the oldest possible Cambrian spore-like packets; older marine rocks with algal cysts	Affinity of the Cambrian packets; any fully reported older sequence	Spore delivery to marine basins, catchment, search effort, spore production by the earliest members

Maximum	Set by	Evidence	Test available	Links assumed
459.1 Ma, living land plants	Hooper et al.	GTS2012 Sandbian base plus uncertainty, keyed to the oldest possible trilete spore (Borenshult-1, Sweden)	Libyan samples reporting no triletes, against records of the same age elsewhere	Earliest crown members produced triletes, the trait persisted, producers lived in the catchment
431.2 Ma, living vascular plants	Hooper et al.	GTS2012 Homerian base plus uncertainty, keyed to Irish Homerian *Cooksonia*	A limit on when the body-fossil record began, and what older ancestors require	Finding rate before the oldest find, delay before recognisable bodies, the oldest finds lie within the vascular crown

Material and methods

(a) Timescale and interpretations. Analysed ages use the stage bases of the International Chronostratigraphic Chart, v2024/12 [25]; literature ages were mapped by stage. Published maxima keep their source timescale, GTS2012, so they can be compared as their authors set them. Each analysis declares which interpretation of disputed identifications it uses, main or inclusive, so contested fossils enter as alternatives.

(b) Source audit. For each maximum we traced the calibration fossil, its age rule and how it was found back to the original descriptions, and recorded each published interpretation of the fossil's identity with its evidence [15]. A reported absence applies to the material examined, and a missing report counts as an absence where the source reports the target fully.

(c) Libyan spore records. Abuhmida [26] reported spores and other microfossils from Ordovician to lower Silurian wells in the Murzuq Basin, southwest Libya. We transcribed all 164 appendix samples with their chart and prose reports. The analysis uses the chart-coded samples of drill cuttings (rock chips returned during drilling) from five wells: three supply 28 samples older than the Llandovery (earliest Silurian), and all five supply 59 Llandovery samples, which serve as the reference. The source states the absence of trilete spores explicitly for part of the sequence; for the Hawaz Formation, samples reporting none rest on complete reporting charts and the stated microscopy protocol. The basin lay near the Ordovician South Pole, within the area of Hirnantian glaciation [27,28] (electronic supplementary material, figure S2). We compared the samples' ages with trilete records in our spore compilation, built from Palynodata [29], the curated assemblages of Wellman et al. [30] and later literature, keeping records with usable ages that are not flagged as reworked or questionable. A model of spore supply, in which samples from one well share a well effect, sets upper limits on how much steady supply the samples allow; its 95% limits were tested on simulated data with known answers (electronic supplementary material, S2).

(d) Body-fossil finds. We assembled land-plant body fossils dated to 453–393 Ma from Paleobiology Database (PBDB) occurrences, whose names an AI model classified with cited sources before our audit, and from a source-reviewed set of the oldest records with their competing interpretations. Collections, taxa or papers sharing a locality and age count as one find, a proxy for an independent fossil horizon. We declare three nested categories. Land-plant bodies are any accepted macro- or mesofossil. Polysporangiophytes have documented branched axes bearing sporangia (spore capsules), or placement within the vascular crown. Crown vascular plants belong to the lycophyte line (clubmosses and relatives) or the euphyllophytes (ferns and seed plants) [31–34]. Under the main interpretation the three categories hold 78, 71 and 50 finds, of which 76, 69 and 48 are certainly older than the end of the Emsian (393.47 Ma), the younger edge of the time window (electronic supplementary material, S5.1).

(e) Limit on when the record began. Let $w(s)$ be a fixed relative rate at which recognisable fossils of

a category are found at age s, and $U(a) = \int_c^a w(s)\,ds$ the cumulative finding rate between a younger cutoff c, here the end of the Emsian, and age a. For n independent finds with oldest age A, the 95% confidence limit T on the start of the recognisable record solves $U(T) = U(A)\,0.05^{-1/n}$ (figure 2). With constant w this is the classical confidence interval on a stratigraphic range, and the absolute rate cancels [23,24]; varying w follows the extensions to unevenly spaced fossil horizons [35,36]. For finds dated by age ranges, A is the oldest possible age of the oldest find, and n counts only finds certainly inside the window; both choices move the limit older. The limit concerns the start of the record; the Discussion treats its link to ancestors.

(f) Finding-rate histories and shape tests. We compared six assumed histories of the finding rate: steady; proportional to PBDB land and marginal-marine collections per Myr [37], with or without the collections that hold plant fossils; declining exponentially into the past with 25 or 10 Myr timescales; and plant-excluded collection density multiplied by the 10 Myr decline. The collection records reach back to 469.4 Ma, the oldest age the analysis can consider. For each history, an exact Kolmogorov–Smirnov test compared the spread of fossil ages it predicts with the observed spread, given the count and the oldest find. Because many ages are known only as ranges, we searched every possible assignment of ages within those ranges, including whether the finds straddling the cutoff fall inside the window, for the best fit [compare 38]. A history that fails under every assignment is rejected (electronic supplementary material, S5.2).

(g) Lower earlier finding rates and delays. The main scenario holds the finding rate steady after 426.7 Ma, the oldest possible age of a crown vascular-plant find, and lowers it before that age. For a candidate ancestor age t and an assumed delay d before the first recognisable bodies, the ratio of earlier to later finding rate that puts $t - d$ where the limit of (e) would fall is $r = (A - c)(0.05^{-1/n} - 1)/(t - d - A)$, defined where $t - d > A$; fossils would then have to be found $1/r$ times less often before the oldest find than after it. Longer delays reduce the reduction a given ancestor age needs [16]. We chose where to put the step after seeing the data, so these curves show what would fit, without the 95% confidence that (e) gives each fixed history.

Results

(a) The 515.5 Ma maximum rests on a qualitative argument. Morris et al. placed this maximum below the oldest possible total-group records, spore-like packets and clusters from the Cambrian Stage 4 Rome Formation, at the base of the *Olenellus* Biozone (about 515.5 Ma). Their preservation argument, from older marine rocks with algal cysts and without land-plant spores, shows that walled microfossils survived and were recovered [1,14]. How many spores reached those basins, the land areas they drained and the search effort remain unmeasured in the sources we assessed. All three sources we examined assign the Cambrian packets to algae or lineages before land plants: Charophyta [39], a pre-embryophytic grade [40] or ancestral aeroterrestrial algae [41] (electronic supplementary material, S4).

(b) The younger maxima follow a stage-base rule. Hooper et al. set both younger maxima as a GTS2012 stage base plus its uncertainty. They keyed 459.1 Ma (458.4 + 0.7 = 459.1 Ma) to a possible trilete spore at 48.2 m in the Borenshult-1 core [21]. The oldest definite trilete level, at 43.1 m, is also Sandbian and lies below the 454.41 Ma Kinnekulle K-bentonite, so the rule gives the same value from either level (electronic supplementary material, S2.4). They keyed 431.2 Ma (430.5 + 0.7 = 431.2 Ma) to Irish Homerian *Cooksonia* [22]. The older Czech *Cooksonia barrandei*, a branched sporophyte with terminal sporangia from a Sheinwoodian (early Silurian) horizon dated from its slab fauna and by correlation [32], meets the same criterion (electronic supplementary material, table S13). On the current timescale the rule gives outer values of 431.9 Ma for the Homerian and 434.1 Ma for the Sheinwoodian.

(c) The Libyan samples show that a spore absence can be local. Trilete spores are reported in 51 of 59 Llandovery samples and in none of the 28 older samples (electronic supplementary material, figure S2). Of the older samples, 18, from three wells, are Katian–Hirnantian (Late Ordovician). They are younger than the definite Swedish triletes below the 454.41 Ma bentonite, and fall in the same stages as trilete-bearing assemblages in Saudi Arabia, Turkey, southern Britain and Belgium [[30]][]; 5 of the Libyan samples, and assemblages in Saudi Arabia and Britain, are Hirnantian, a stage of about two million years. Trilete spores thus occur elsewhere in the stages these Libyan rocks represent, and none is reported from them. Only 10 broadly dated Darriwilian (Middle Ordovician) samples, from two wells, have age ranges reaching beyond 459.1 Ma, and their weight depends on how ages are spread within the stage. With equal delivery, the samples limit steady trilete supply since 459.1 Ma to 0.091 of the Llandovery level under the main model, and to

about 0.4 when samples are weighted by well and rock unit (electronic supplementary material, S2.2 and S2.5). Lower delivery, lower local supply or producers outside the catchment all fit.

(d) At a steady finding rate, the 95% confidence limit on the vascular-plant body record's start is near 429 Ma. If crown vascular-plant fossils were found at a steady rate, the 48 finds certainly inside the window put the 95% confidence limit on the start of their record at 428.8 Ma (figure 3*a*). Finding rates that track collection density give 428.9 Ma, or 428.5 Ma with plant-bearing collections removed, and a 25 Myr decline into the past gives 431.6 Ma. If finding fossils became steeply harder with age (the 10 Myr decline), the method sets no limit as far back as 469.4 Ma, the edge of our data. That history, however, predicts a spread of fossil ages unlike the one observed, however the uncertain ages are assigned. The 25 Myr decline passes the shape test under one assignment of ages when all 50 possible finds are counted ($p = 0.067$), and fails under every assignment of the 48 finds certainly inside the window (best $p = 0.035$).

(e) Counting fewer independent finds moves the limit past 431.2 Ma. Treating each region and stage as one find (19 finds) gives 432.4 Ma, and counting only the curated oldest records (15 finds) gives 434.05 Ma, both at a steady finding rate. Shifting stage boundaries by their quoted uncertainties, in the scenarios tested, moves the main limit between 427.31 and 430.42 Ma (electronic supplementary material, S5.3).

(f) Older vascular-plant dates fit if early fossils were found less often, or recognisable bodies appeared later. In the main scenario, halving the finding rate before the oldest find moves the limit to 431.0 Ma, still younger than the 431.2 Ma maximum. Ancestors at 444–452 Ma, spanning the Morris et al. mean dates and the Harris et al. interval, fit if fossils were found 8–12 times less often before the oldest find than after it, with recognisable bodies appearing with the ancestor (figure 3*b*). Counting by region and stage, or only the curated oldest records, 2.4–4.5 times suffices, and 431.2 Ma already fits a steady rate. At a steady finding rate, the same ancestors fit if recognisable bodies appeared at least 15–23 Myr after them. Smaller reductions combine with shorter delays (figure 3*b*).

(g) One Ordovician find moves the broader land-plant limit. Sporangial fragments with in situ cryptospores from the Ghaba-1 borehole, Oman, dated to the Katian on later correlation [42], form one land-plant body find. Adding them moves the steady-rate limit for all land-plant body fossils from 434.5 to 455.2 Ma, and the enlarged set fails the shape test under all six histories. The fragments came from palynological preparation of borehole material, whereas most later finds are compressions collected from bedding planes [43].

Discussion

The early record allows older ancestors, and the analyses state what each would require. The evidence differs in depth between the two clades. For living land plants it comes from the audit of the 515.5 Ma maximum and one polar basin, whose samples report no triletes from stages in which triletes occur elsewhere. Those samples exclude a steady, well-delivered supply near the Llandovery level, while lower delivery, lower local supply or spore producers outside the catchment all fit. For living vascular plants it comes from a body record drawn from several regions: at a steady finding rate, or one tracking collection density, the 95% confidence limit on the start of the recognisable crown vascular-plant record lies near 429 Ma, and counting fewer independent finds puts it at 432.4–434.05 Ma. For each way of counting, a lower earlier finding rate, later recognisable bodies or both fit ancestors older than these limits; the models give the size of each requirement in aggregate.

Several processes could have made early plants scarcer in the record. Producers may have been rarer, more geographically restricted, concentrated where little terrestrial sediment accumulated, or preserved in intervals that have been searched less. Body size, cuticle and lignified tissue could also raise the chance that a body is preserved, found and recognised. Branched sporophytes and conducting tissue were acquired stepwise along the lineages being dated [44]. The early branched sporophyte *C. barrandei* is preserved in marine sediments [32], and the Oman fragments show that small sporangia can survive [42]. The models give the combined reduction a candidate age requires; the contribution and plausibility of each process remain open.

Silvestro et al. [45] modelled genus-level PBDB plant occurrences with a joint Bayesian model of preservation and diversification, and estimated the origin of vascular plants at a mean of 433.5 Ma (95% highest posterior density 449.0–424.0 Ma). They estimate when the earliest vascular-plant genus lineage arose, with a fitted,

varying preservation rate; we limit the start of the recognisable record of a declared crown category under assumed finding-rate histories, then ask what older ancestors require. The two analyses target different quantities from partly shared data and test different assumptions. CladeDate builds calibration densities directly from fossil times [46], and Hooper et al. link spore-character reconstruction to revised calibrations. We make the maxima's observational requirements explicit, limiting local spore supply and sizing the reductions in finding rate, or the delays, that fit older vascular-plant dates.

Conditions. The Libyan limits assume that the Llandovery spores are indigenous, that older and younger samples were reported comparably, that ages are uniform within stages, and that samples are independent apart from a shared well effect; the four-tenths figure comes from an exploratory weighting that was not tested on simulated data (electronic supplementary material, S2.5). The comparison with other regions holds at stage resolution and depends on the coverage of the spore compilation. Linking the local spore record to ancestors requires that the earliest crown embryophytes produced recognisable trilete spores, that the trait persisted and that producers lived in the catchment (electronic supplementary material, S6).

The body-fossil limits and shape rejections assume independent ages on a continuous scale; fixed finding-rate histories, cutoff, timescale, inventory and counting rules; and an observation process whose reported age ranges always contain the true ages, including whether each find falls inside the window. Geological age ranges carry no demonstrated joint coverage; if the probability that any range misses its true age is δ, the test's error rate is at most $\alpha + \delta$, where α is its level for exact ages. Simulations with exact ages gave coverage of 0.941–0.958, falling to about 0.78 when paired duplicate horizons were counted as independent (electronic supplementary material, S5.6). The step scenario was chosen after seeing the data. Linking the vascular-plant limit to ancestors requires that the oldest finds belong to the crown group, and a specified delay before recognisable bodies.

Published evidence could narrow these requirements. Records of searched Silurian terrestrial beds, with effort and empty beds included, would help constrain early finding rates, and compilations of rock units could supply proxies for the terrestrial rock available at each age [47]. A second complete spore sequence outside polar Gondwana, with older samples reporting no triletes, younger samples reporting them and counted target and control grains, would show whether supply limits of this size hold outside one polar basin; a fully reported marine sequence older than 515.5 Ma, with comparable recovery documented, would help quantify the oldest maximum's absence argument. Finer, independent ages for the Yea (Victoria, Australia) and Bathurst Island (Arctic Canada) horizons, which hold the oldest possible crown vascular-plant finds, would sharpen the body-fossil limit [31,33]. Characters placing the oldest branched sporophytes on the lycophyte or euphyllophyte side of the vascular crown could change which fossils enter the category, as well as their link to the ancestor.

Conclusion

A fossil maximum based on absence combines expectations about ancestral characters with the chance of finding their remains in older rocks. Clock builders can state each maximum's character, geographical and finding-rate assumptions, and justify the probability assigned to older ages against them [12,15]. Testing alternative defensible maxima then shows how much each assumption moves the timescale, and sampling the joint age prior without molecular data, then comparing its effective marginal ages with the posteriors, shows what the molecular data add [48]. The same separation of the evidence, the chance of finding it and its link to the ancestor applies to any maximum argued from absence. The Libyan basin shows that trilete spores can go unreported in rocks from stages in which they occur elsewhere. Older land-plant ancestors fit its record if, with equal delivery, their steady trilete supply since 459.1 Ma stayed below a tenth of the later supply (0.091; four-tenths under an exploratory weighting), or their spores never reached the basin, because the producers lived elsewhere or triletes evolved later. Vascular-plant ancestors at 444–452 Ma fit if early body fossils were found 8–12 times less often, with recognisable bodies appearing with the ancestor, or, at a steady rate, if those bodies appeared at least 15–23 Myr after it; the first is a scenario chosen after seeing the data. Missing fossils thus allow older ancestors if early plants were found less often or became recognisable later, and the records examined state how much less often.

Ethics

We analysed published descriptions and data. The study involved no fieldwork, specimen collection or research involving human participants.

Data accessibility

The repository https://github.com/BWBrook/silurian_plants holds the source-linked derived tables, model code, qualification evidence, body-record outputs with hash receipts, figure and table scripts, and the claims ledger that binds every reported number to its evidence. The frozen PBDB collections extract used by the effort histories is archived under CC0 [37]. Copyrighted full texts are excluded from the repository; source routes and held-file hashes are recorded separately. Openly licensed third-party deposits are fetched by script and hash-checked. Supplementary material is available online.

Declaration of AI use

Claude Opus 5.5 (Anthropic) developed and ran analyses and drafted and revised the manuscript. OpenAI Codex (`gpt-6.1-sol` and `gpt-6.1-terra`) curated and checked source data, implemented analyses, audited statistical methods and reproduction, and revised the manuscript. Earlier Codex work used `gpt-5.6-sol` and `gpt-5.6-luna` for source recovery and methodological checks. Claude and Codex reviewed each other's work. GPT-6 Pro (OpenAI) supplied independent critiques, which were checked against sources and saved outputs before integration. Barry W. Brook originated and guided the programme and contributed scientific decisions; he will review the final manuscript before submission. Task-specific roles, model versions and the limits of sessions whose exact variant was unavailable are recorded in `docs/charter.md`, `docs/audits/` and `manuscript/reconstruction/`, with adopted decisions in `docs/decisions.md`.

Authors' contributions

[Byline and final CRediT wording remain deferred to the submission phase. A contribution inventory is prepared in `manuscript/reconstruction/4-submission-materials.md`.]

Conflict of interest declaration

[Barry to confirm.]

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References

1. Morris JL *et al.* 2018 The timescale of early land plant evolution. *Proceedings of the National Academy of Sciences* **115**. (doi:10.1073/pnas.1719588115)

2. Harris BJ, Clark JW, Schrempf D, Szöllősi GJ, Donoghue PCJ, Hetherington AM, Williams TA. 2022 Divergent evolutionary trajectories of bryophytes and tracheophytes from a complex common ancestor of land plants. *Nature Ecology & Evolution* **6**, 1634–1643. (doi:10.1038/s41559-022-01885-x)

3. Hooper J, Worth AM, Wellman CH, Clark JW, Donoghue PCJ. 2026 Reconstructing spore evolution recalibrates the timescale of land plant diversification. *Nature Plants* (doi:10.1038/s41477-026-02432-3)

4. Su D, Yang L, Shi X, Ma X, Zhou X, Hedges SB, Zhong B. 2021 Large-Scale Phylogenomic Analyses Reveal the Monophyly of Bryophytes and Neoproterozoic Origin of Land Plants. *Molecular Biology and Evolution* **38**, 3332–3344. (doi:10.1093/molbev/msab106)

5. Kenrick P, Wellman CH, Schneider H, Edgecombe GD. 2012 A timeline for terrestrialization: consequences for the carbon cycle in the Palaeozoic. *Philosophical Transactions of the Royal Society B: Biological Sciences* **367**, 519–536. (doi:10.1098/rstb.2011.0271)

6. Donoghue PCJ, Harrison CJ, Paps J, Schneider H. 2021 The evolutionary emergence of land plants. *Current Biology* **31**, R1281–R1298. (doi:10.1016/j.cub.2021.07.038)

7. Strother PK, Foster C. 2021 A fossil record of land plant origins from charophyte algae. *Science* **373**, 792–796. (doi:10.1126/science.abj2927)

8. Lenton TM, Crouch M, Johnson M, Pires N, Dolan L. 2012 First plants cooled the Ordovician. *Nature Geoscience* **5**, 86–89. (doi:10.1038/ngeo1390)

9. Benton MJ, Donoghue PCJ. 2006 Paleontological Evidence to Date the Tree of Life. *Molecular Biology and Evolution* **24**, 26–53. (doi:10.1093/molbev/msl150)

10. Ho SYW, Phillips MJ. 2009 Accounting for Calibration Uncertainty in Phylogenetic Estimation of Evolutionary Divergence Times. *Systematic Biology* **58**, 367–380. (doi:10.1093/sysbio/syp035)

11. dos Reis M, Donoghue PCJ, Yang Z. 2015 Bayesian molecular clock dating of species divergences in the genomics era. *Nature Reviews Genetics* **17**, 71–80. (doi:10.1038/nrg.2015.8)

12. Yang Z, Rannala B. 2005 Bayesian Estimation of Species Divergence Times Under a Molecular Clock Using Multiple Fossil Calibrations with Soft Bounds. *Molecular Biology and Evolution* **23**, 212–226. (doi:10.1093/molbev/msj024)

13. Hedges SB, Tao Q, Walker M, Kumar S. 2018 Accurate timetrees require accurate calibrations. *Proceedings of the National Academy of Sciences* **115**. (doi:10.1073/pnas.1812558115)

14. Morris JL *et al.* 2018 Reply to Hedges et al.: Accurate timetrees do indeed require accurate calibrations. *Proceedings of the National Academy of Sciences* **115**. (doi:10.1073/pnas.1812816115)

15. Parham JF *et al.* 2011 Best Practices for Justifying Fossil Calibrations. *Systematic Biology* **61**, 346–359. (doi:10.1093/sysbio/syr107)

289 16. Marshall CR. 2019 Using the Fossil Record to Evaluate Timetree Timescales. *Frontiers in Genetics* **10**. (doi:10.3389/fgene.2019.01049)

290 17. Wellman CH, Gray J. 2000 The microfossil record of early land plants. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences* **355**, 717–732. (doi:10.1098/rstb.2000.0612)

291 18. Rubinstein CV, Gerrienne P, de la Puente GS, Astini RA, Steemans P. 2010 Early Middle Ordovician evidence for land plants in Argentina (eastern Gondwana). *New Phytologist* **188**, 365–369. (doi:10.1111/j.1469-8137.2010.03433.x)

292 19. Steemans P, Hérissé AL, Melvin J, Miller MA, Paris F, Verniers J, Wellman CH. 2009 Origin and Radiation of the Earliest Vascular Land Plants. *Science* **324**, 353–353. (doi:10.1126/science.1169659)

293 20. Edwards D, Morris JL, Richardson JB, Kenrick P. 2014 Cryptospores and cryptophytes reveal hidden diversity in early land floras. *New Phytologist* **202**, 50–78. (doi:10.1111/nph.12645)

294 21. Rubinstein CV, Vajda V. 2019 Baltica cradle of early land plants? Oldest record of trilete spores and diverse cryptospore assemblages; evidence from Ordovician successions of Sweden. *GFF* **141**, 181–190. (doi:10.1080/11035897.2019.1636860)

295 22. Edwards D, Feehan J. 1980 Records of Cooksonia-type sporangia from late Wenlock strata in Ireland. *Nature* **287**, 41–42. (doi:10.1038/287041a0)

296 23. Strauss D, Sadler PM. 1989 Classical confidence intervals and Bayesian probability estimates for ends of local taxon ranges. *Mathematical Geology* **21**, 411–427. (doi:10.1007/bf00897326)

297 24. Marshall CR. 1990 Confidence intervals on stratigraphic ranges. *Paleobiology* **16**, 1–10. (doi:10.1017/s0094837300009672)

298 25. Cohen KM, Finney SC, Gibbard PL, Fan J-X. 2013 The ICS International Chronostratigraphic Chart. *Episodes* **36**, 199–204. (doi:10.18814/epiiugs/2013/v36i3/002)

299 26. Abuhmida FH. 2013 Palynological analysis of the Ordovician to Lower Silurian sediments from the Murzuq Basin, southwest Libya. PhD thesis, University of Sheffield. See https://etheses.whiterose.ac.uk/id/eprint/4475/.

300 27. Cocks LRM, Torsvik TH. 2021 Ordovician palaeogeography and climate change. *Gondwana Research* **100**, 53–72. (doi:10.1016/j.gr.2020.09.008)

301 28. Scotese CR, Song H, Mills BJW, van der Meer DG. 2021 Phanerozoic paleotemperatures: The earth's changing climate during the last 540 million years. *Earth-Science Reviews* **215**, 103503. (doi:10.1016/j.earscirev.2021.103503)

302 29. White JM. 2008 Palynodata Datafile: 2006 version. (doi:10.4095/225704)

303 30. Wellman CH, Steemans P, Vecoli M. 2013 Chapter 29 Palaeophytogeography of Ordovician–Silurian land plants. *Geological Society, London, Memoirs* **38**, 461–476. (doi:10.1144/m38.29)

304 31. Kotyk ME, Basinger JF, Gensel PG, de Freitas TA. 2002 Morphologically complex plant macrofossils from the Late Silurian of Arctic Canada. *American Journal of Botany* **89**, 1004–1013. (doi:10.3732/ajb.89.6.1004)

305 32. Libertín M, Kvaček J, Bek J, Žárský V, Štorch P. 2018 Sporophytes of polysporangiate land plants from the early Silurian period may have been photosynthetically autonomous. *Nature Plants* **4**, 269–271. (doi:10.1038/s41477-018-0140-y)

306 33. Earp C. 2024 Notanopliid brachiopods and associated fossils from the Yea area, central Victoria: a key to resolving age and correlation problems in a complex geological setting. *Proceedings of the Royal Society of Victoria* **136**. (doi:10.1071/rs23021)

307 34. Libertín M, Kvaček J, Bek J. 2024 The genus Aberlemnia and its Silurian–Devonian fossil record. *Review of Palaeobotany and Palynology* **320**, 105017. (doi:10.1016/j.revpalbo.2023.105017)

308 35. Marshall CR. 1994 Confidence intervals on stratigraphic ranges: partial relaxation of the assumption of randomly distributed fossil horizons. *Paleobiology* **20**, 459–469. (doi:10.1017/s0094837300012938)

309 36. Marshall CR. 1997 Confidence intervals on stratigraphic ranges with nonrandom distributions of fossil horizons. *Paleobiology* **23**, 165–173. (doi:10.1017/s0094837300016766)

310 37. Brook BW. 2026 Frozen Paleobiology Database all-taxa collections extract: Darriwilian–Emsian, 28 October 2026. (doi:10.5281/zenodo.24028289)

311 38. Berger RL, Boos DD. 1994 *P* Values Maximized Over a Confidence Set for the Nuisance Parameter. *Journal of the American Statistical Association* **89**, 1012–1016. (doi:10.1080/01621459.1994.10476836)

312 39. Strother PK. 2016 Systematics and evolutionary significance of some new cryptospores from the Cambrian of eastern Tennessee, USA. *Review of Palaeobotany and Palynology* **227**, 28–41. (doi:10.1016/j.revpalbo.2015.10.006)

313 40. Taylor WA, Strother PK. 2024 Ultrastructure of Cambrian cryptospores and the early evolution of the plant spore wall. *Fossil Imprint* **80**, 90–106. (doi:10.37520/fi.2024.009)

314 41. Wang K, Xu H-H, Yin L-M. 2022 A palynological assemblage from the Cambrian (Series 2, Stage 4) of Shandong Province, China, and its implications to the transition from algae to land plants. *Review of Palaeobotany and Palynology* **301**, 104645. (doi:10.1016/j.revpalbo.2022.104645)

315 42. Wellman CH, Osterloff PL, Mohiuddin U. 2003 Fragments of the earliest land plants. *Nature* **425**, 282–285. (doi:10.1038/nature01884)

316 43. Edwards D, Rogerson ECW. 1979 New records of fertile Rhyniophytina from the late Silurian of Wales. *Geological Magazine* **116**, 93–98. (doi:10.1017/s0016756800042503)

317 44. Kenrick P, Crane PR. 1997 The origin and early evolution of plants on land. *Nature* **389**, 33–39. (doi:10.1038/37918)

318 45. Silvestro D, Cascales-Miñana B, Bacon CD, Antonelli A. 2015 Revisiting the origin and diversification of vascular plants through a comprehensive Bayesian analysis of the fossil record. *New Phytologist* **207**, 425–436. (doi:10.1111/nph.13247)

319 46. Claramunt S. 2022 CladeDate : Calibration information generator for divergence time estimation. *Methods in Ecology and Evolution* **13**, 2331–2338. (doi:10.1111/2041-210x.13977)

320 47. Peters SE, Husson JM, Czaplewski J. 2018 Macrostrat: A Platform for Geological Data Integration and Deep-Time Earth Crust Research. *Geochemistry, Geophysics, Geosystems* **19**, 1393–1409. (doi:10.1029/2018gc007467)

48. Brown JW, Smith SA. 2017 The Past Sure is Tense: On Interpreting Phylogenetic Divergence Time Estimates. *Systematic Biology* **67**, 340–353. (doi:10.1093/sysbio/syx074)

322 **Fig. 1.**

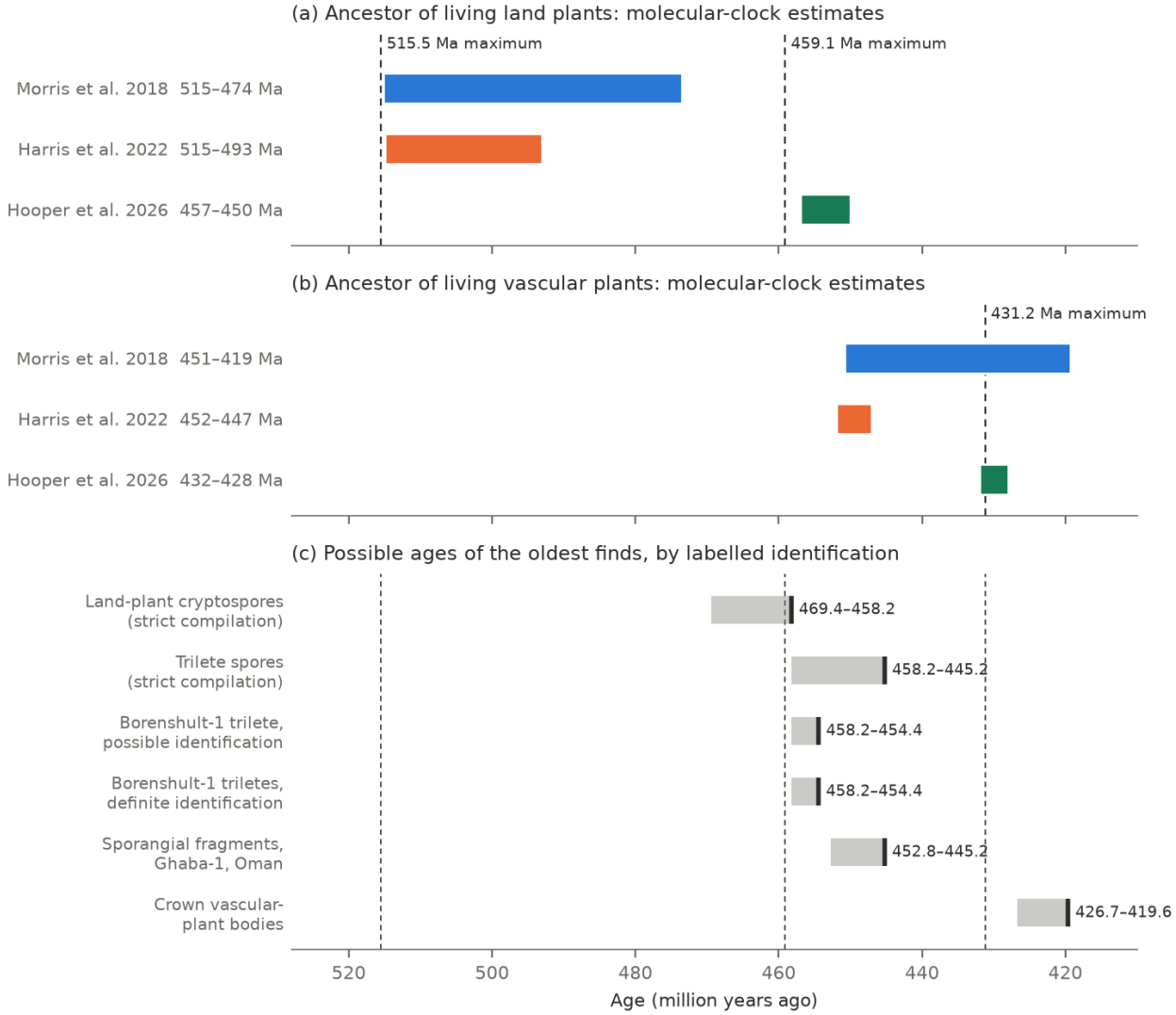


Figure 1: The dating dispute and its fossil evidence. Upper bars give dates for the ancestor of living land plants; the next group gives living vascular plants. Morris et al.'s bars envelope 95% highest posterior density (HPD) intervals across seven tree topologies; Harris et al. and Hooper et al. give their study-specific 95% HPD intervals [1–3]. Vertical lines mark the maxima of 515.5, 459.1 and 431.2 Ma. Clock estimates and maxima keep their published numerical ages. Fossil stage ages use the International Chronostratigraphic Chart v2024/12; independent radiometric ages keep their published values. Each fossil bar spans the possible ages of a category's oldest find, from earliest possible to earliest certain occurrence; the two ends may come from different finds. Separate bars show cryptospores and triletes under the strict compilation interpretation, possible and definite Borenshult triletes [21], Oman's sporangial fragments [42], and crown vascular-plant body fossils [31,33]. The two Borenshult interpretations share this coarse age bracket. Each age holds for the identification labelled; the queried specimen remains a possible trilete.

Fig. 2.

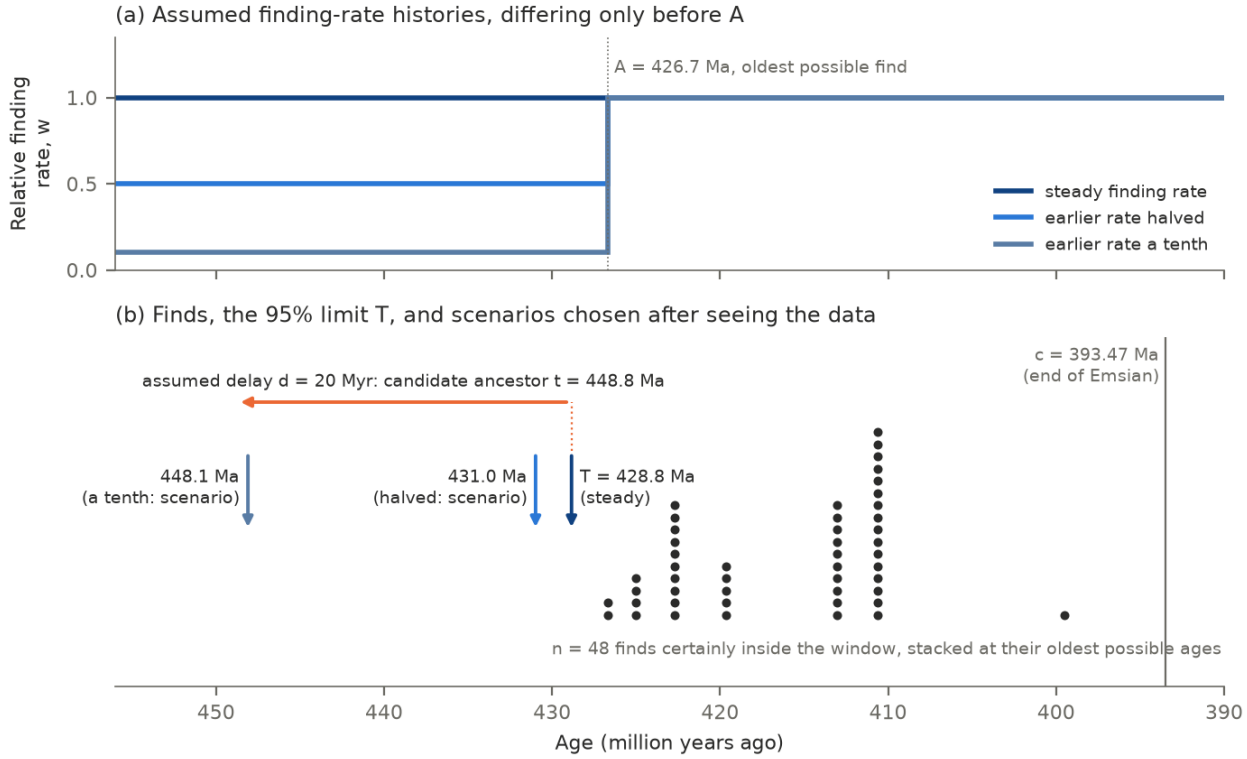


Figure 2: How the body-fossil limit links fossils to ancestors, shown with the main crown vascular-plant finds. (*a*) The finding rate is steady after the oldest find; before it, the rate is unchanged, halved or cut to a tenth. (*b*) Dots show the 48 finds certainly inside the window, at the oldest ends of their age ranges and stacked for visibility. For an assumed finding-rate history, the oldest find A and the count n give the 95% confidence limit T on the start of the recognisable record. For finds dated by age ranges, A is the oldest possible age of the oldest find and n counts only finds certainly inside the window. In the scenarios chosen after seeing the data, halving the earlier finding rate gives a compatibility endpoint of 431.0 Ma, and cutting it to a tenth gives 448.1 Ma; the steady-rate 95% confidence limit is 428.8 Ma. An assumed delay d before recognisable bodies links the start of the record to a candidate ancestor age t; the arrow adds a 20 Myr delay to the steady-rate limit, giving 448.8 Ma.

Fig. 3.

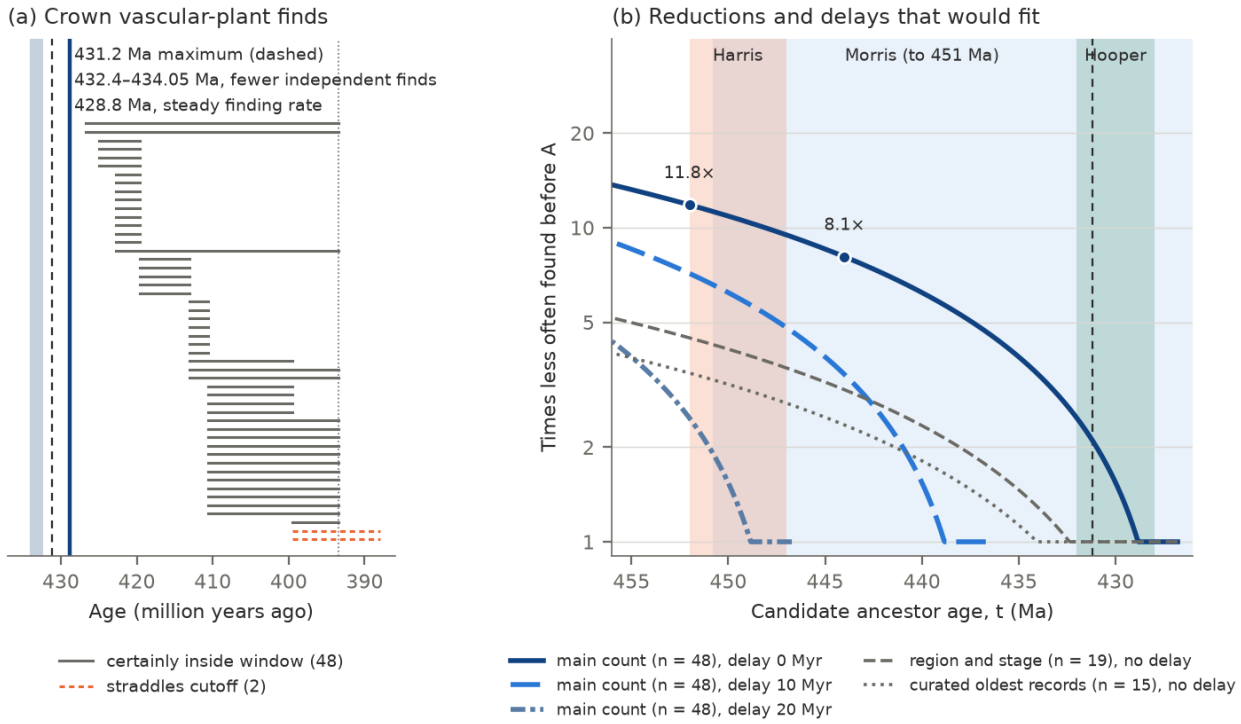


Figure 3: What older vascular-plant ancestors require of the fossil record. (*a*) The 50 possible locality–age finds, ordered by oldest possible age; 48 lie certainly inside the window, and the two straddling the cutoff are marked separately. At a steady finding rate the 95% confidence limit is 428.8 Ma; counting fewer independent finds, by region and stage or only the curated oldest records, gives 432.4–434.05 Ma. (*b*) How many times less often fossils would have to have been found before the oldest find than after it, for each candidate ancestor age t. The finding rate is steady after $A = 426.7$ Ma, the oldest possible age of the oldest find, and the younger cutoff is $c = 393.47$ Ma. Where a curve rises above one, an oldest find no older than A has probability 0.05 given the count, the rule behind the steady-rate limit, with the record starting at $t - d$. Main curves use $n = 48$ and assumed delays d of zero, ten and twenty Myr before recognisable bodies; zero-delay comparison curves count by region and stage ($n = 19$) or only the curated oldest records ($n = 15$). Shading marks the published clock intervals; Morris's younger interval continues outside this panel (figure 1). Combinations with $t - d \leq A$ are omitted; flat portions at one show ages a steady finding rate already allows. We placed the step after seeing the data, so the curves show what would fit and carry no 95% confidence. Older ancestors fit if early fossils were found several times less often, or recognisable bodies appeared later.